

DAVID BAEK

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SUMMARY

MIT Ph.D. student focusing on building reliable and safe AI systems, with extensive experience in **LLM post-training** and **Reinforcement Learning** for both academic research and industrial applications. My research has been published in top-tier conferences (NeurIPS, ICML, and ICLR). I am passionate about building scalable, efficient, and safe AI systems that can positively impact our society.

EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, MA, USA

Ph.D. in Electrical Engineering & Computer Science (EECS), GPA: 5.0/5.0

Sep 2023 – May 2027 (Expected)

- Advisor: Max Tegmark
- Research Area: Responsible AI, Large Language Models, AI Alignment

Seoul National University (SNU)

Seoul, Korea

B.S. in Physics and Computer Science, Summa Cum Laude, GPA: 4.23/4.3

Mar 2017 – Aug 2023

- Presidential Award (Ranked **1st** among graduating cohort in College of Natural Sciences)
- Includes two years on leave for compulsory military service (2020–21, Job: Cyber Security Specialist)

SELECTED PUBLICATIONS

1. **[ICML'26 Spotlight]** David Baek, Xinnuo Li, Anay Gupta, Taslim Mahbub, Max Tegmark, and Shi Feng, “Sycophancy Towards Researchers Drives Performative Misalignment.”
2. **[ICML'26]** David Baek*, Andrew Estornell*, Yichi Zhang*, Muhammad Faaiz Taufiq, Jean-Francois Ton, Jie Mei, and Tao Wang, “D-FUSER: Diverse Failure, Unified Success via Error Distribution Shaping in LLM Reasoning.”
3. **[ICLR'26]** Jiawei Zhang, Andrew Estornell, David Baek, Bo Li, and Xiaojun Xu, “Any-Depth Alignment: Unlocking Innate Safety Alignment of LLMs to Any-Depth.”
4. **[NeurIPS'25 Spotlight]** David Baek*, Joshua Engels*, Subhash Kantamneni*, and Max Tegmark, “Scaling Laws for Scalable Oversight.”

EXPERIENCE

Google

May 2026 - Aug 2026

Software Engineer Intern (PhD) (Mentor: Jiaqi Pan)

Mountain View, CA, USA

- Research on Reinforcement Learning and Reward Hacking in LLMs

ByteDance

Jun 2025 - Aug 2025

Machine Learning Research/Engineer Intern (Mentor: Jie Mei, Andrew Estornell)

Bellevue, WA, USA

- Research on Multi-agent Reinforcement Learning and improving the safety and robustness of LLMs
- Proposed a novel product moderation paradigm in TikTok Shop and trained a 7B model, saving \$660k/year in total

Tegmark AI Safety Group

Dec 2023 - Present

Graduate Research Assistant (Advisor: Prof. Max Tegmark)

Cambridge, MA, USA

- Studied geometrical structure of representations in LLMs, with extensive experience in fine-tuning/training LLMs

HONORS & AWARDS (SELECTED)

- Gold Level Certificate, WorldQuant Brain Platform, 2024
- Silver Medal, University Physics Competition, 2018
- Finalist, Samsung Collegiate Programming Cup (SCPC), 2018
- Silver Medal, Korean Mathematical Olympiad (High School Division), 2016
- Silver Medal, International Junior Science Olympiad (IJSO), 2014

TECHNICAL SKILLS

Mathematics: Probability, Statistics, Stochastic Processes, Time Series Analysis, Linear Algebra, Optimization

Programming: Python, C/C++, Java, Matlab, Mathematica, $\LaTeX$, HTML, Javascript

Libraries: PyTorch, Tensorflow[†], HuggingFace, Wandb, Numpy, Scipy, QuTiP, Vue.js/Vuetify, etc.

Machine Learning: Large Language Models, Diffusion Models, Computer Vision, Interpretability Techniques

COMMUNITY SERVICE/OTHER INFORMATION

- Chair of Publicity & Communications Committee @ MIT Ashdown House

Nov 2023 - Present

- Vice President of Publicity @ MIT EECS Graduate Student Association

Jan 2024 - Jan 2025

- Undergraduate Student Research Mentoring: Riya Tyagi (Spring 2025), Duru Ozer (Spring 2025)